

Supplement D — Extended Distributed Reorganization in Coupled Map Lattices

D1. Purpose and Scope

This supplement extends the distributed organization analysis introduced in the main manuscript. The objective is to provide a more detailed characterization of distributed organization behavior in coupled map lattice (CLM) dynamics, with particular emphasis on epsilon-star localization, cluster persistence, robustness behavior, and distributed organization collapse under shuffle controls.

This supplement extends the CLM analysis presented in the main manuscript and should be read alongside Supplement B.

D2. CLM Architecture and Distributed Dynamics

The coupled map lattice regime extends the framework from low-dimensional trajectory organization toward distributed collective organization. The epsilon coupling parameter controls the transition between fragmented local dynamics, intermediate organization regimes, and rigid synchronization. Intermediate epsilon regions near $\epsilon \approx 0.18$ consistently displayed the strongest distributed organization signatures (*see Figures D1 and D4*).

D3. Epsilon-Star Localization

The epsilon-star region was evaluated across multiple scans and parameter sweeps. The observed localization near $\epsilon \approx 0.18$ remained stable under repeated evaluations and appeared as a distributed organization regime rather than a sharp bifurcation-like transition. The distributed organization peak was associated with increased persistence and reduced switching activity (*see Figures D1 and D2*).

D4. Bootstrap Stability

Bootstrap analyses were used to estimate the stability of the distributed organization window. Repeated resampling preserved the localization of the main epsilon-star region, supporting the interpretation that the observed signatures were not single-run artifacts. The bootstrap layer is interpreted as a robustness indicator rather than a formal proof of universality (*see Figure D2*).

D5. Shuffle and Null Controls

Shuffle randomization and null procedures weakened persistence-related structures and distributed organization signatures. The suppression was selective rather than uniform across the parameter range, suggesting that the distributed organization behavior depends on relational organization rather than static geometric properties alone (*see Figures D3 and D9*).

D6. Cluster Persistence and Switching Structure

Cluster persistence analyses demonstrated increased dwell stability and reduced switching behavior near the principal distributed organization region. The resulting organization patterns support the interpretation of localized distributed stabilization under compatibility-dependent constraints (*see Figures D5 and D8*).

D7. Endpoint and Normalization Robustness

Additional scans examined endpoint exclusion behavior and normalization sensitivity. Cross-system comparisons remain operational and exploratory rather than physically equivalent. The objective of the framework is to identify recurring organizational signatures rather than derive universal quantitative scaling laws (*see Figures D6 and D9*).

D8. Interpretation Boundaries

The supplement does not claim universal equivalence between nonlinear systems. The presented evidence should be interpreted as comparative support for recurring distributed organization signatures under compatibility-dependent constraints. Independent replication and future statistical refinement remain necessary.

D9. Quantitative Comparative Summary

Table D1. Comparative distributed organization metrics across coupled map lattice dynamics.

Metric	Observed behavior	Robustness response	Interpretation
ϵ -star region	Stable localization near $\epsilon \approx 0.18$	Stable under bootstrap	Distributed organization regime
Cluster persistence	Increased dwell stability	Reduced after shuffle	Localized stabilization
Switching activity	Suppressed near principal $\epsilon \approx 0.18$ region	Recovered after randomization	Compatibility-dependent organization
Distributed organization	Enhanced intermediate coupling	Selective collapse under controls	Collective organization

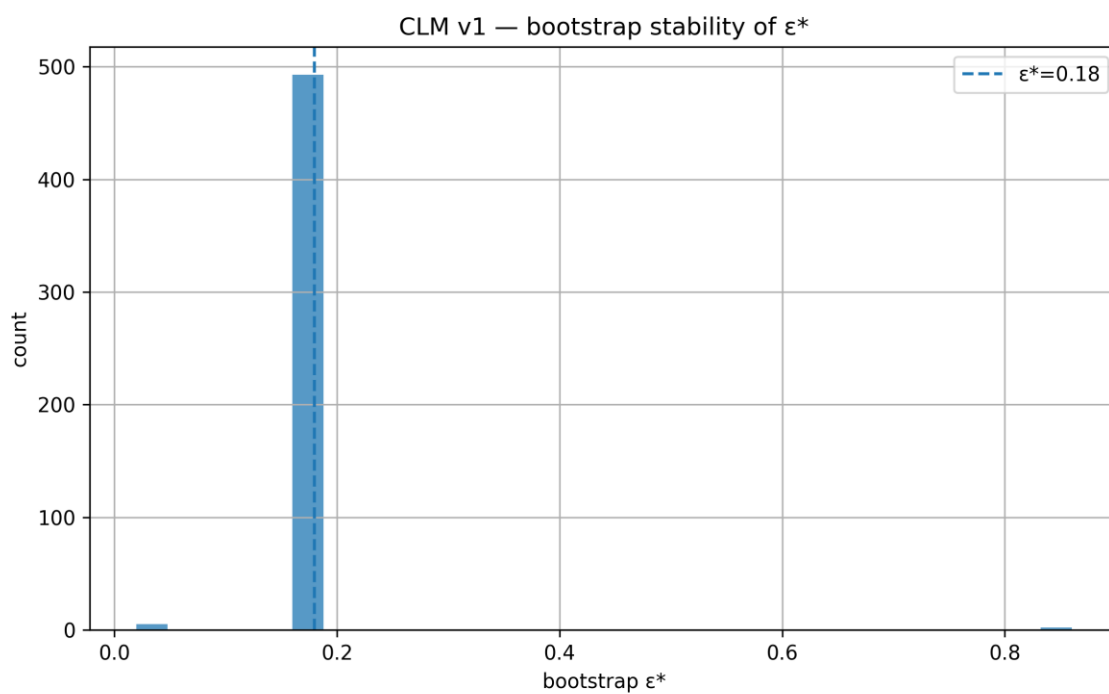


Figure D1. Localization of the primary epsilon-star distributed organization region.

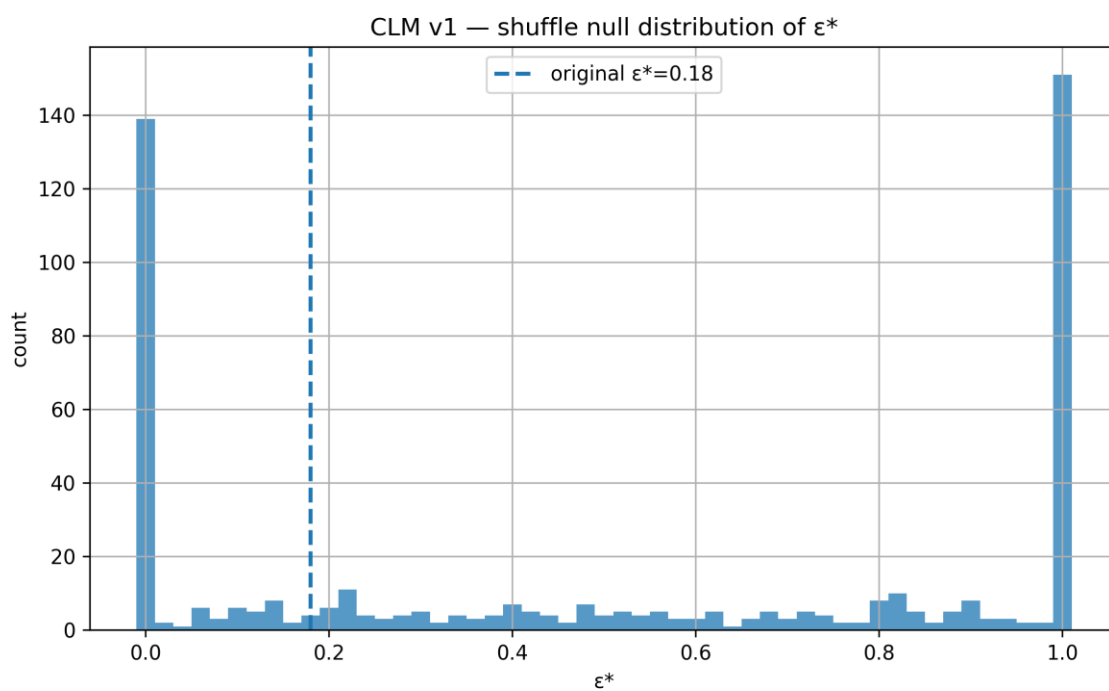


Figure D2. Bootstrap stability of the epsilon-star window across repeated resampled realizations.

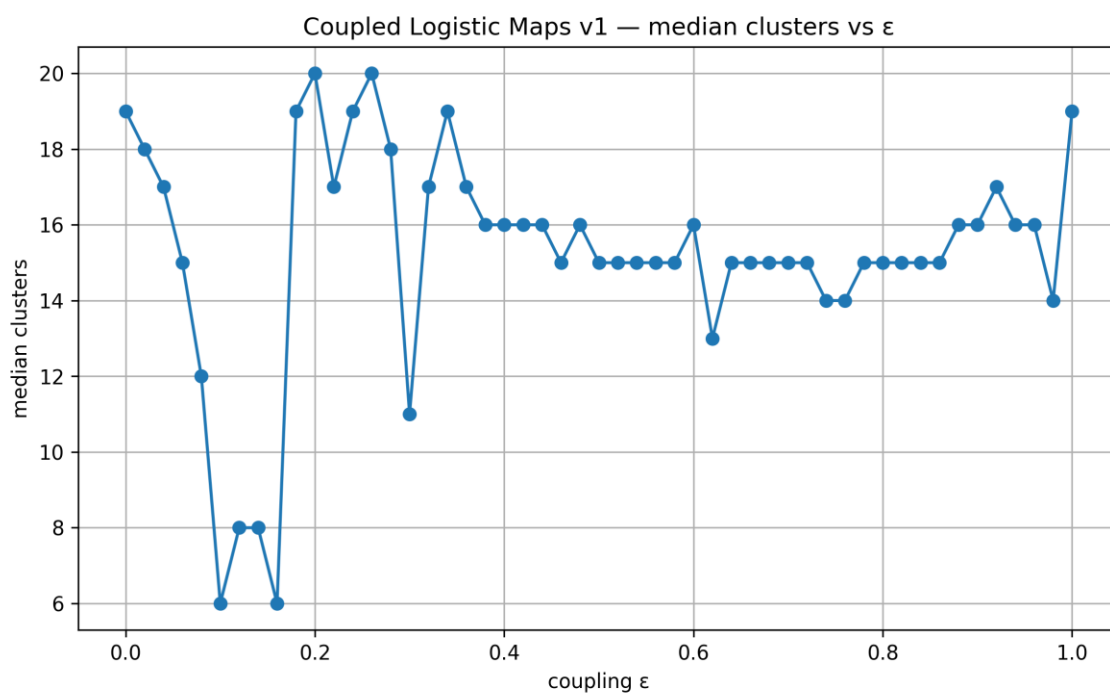


Figure D3. Distributed persistence structure before and after shuffle randomization.

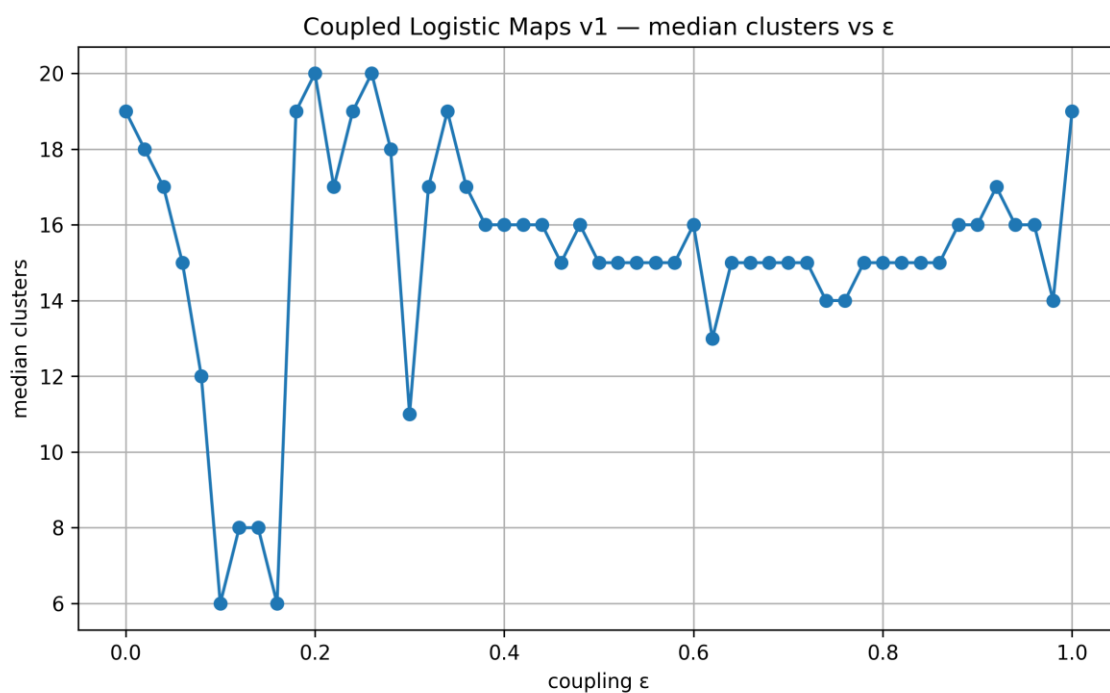


Figure D4. Representative distributed organization overlay in the coupled map lattice regime.

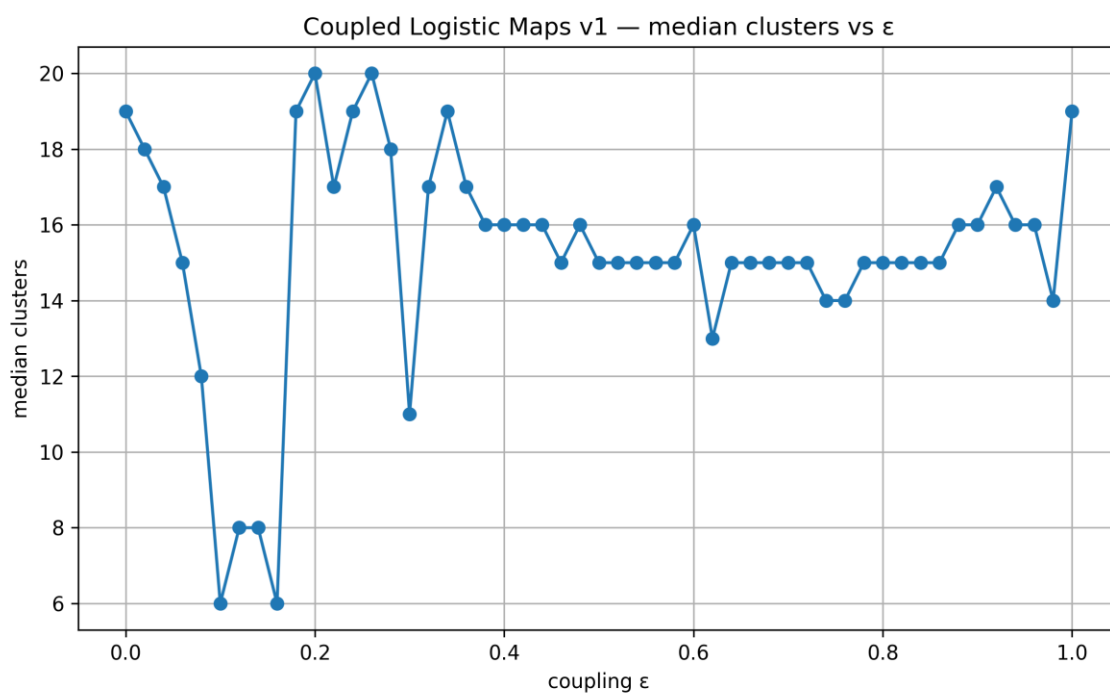


Figure D5. Cluster persistence and switching suppression near the principal organization window.

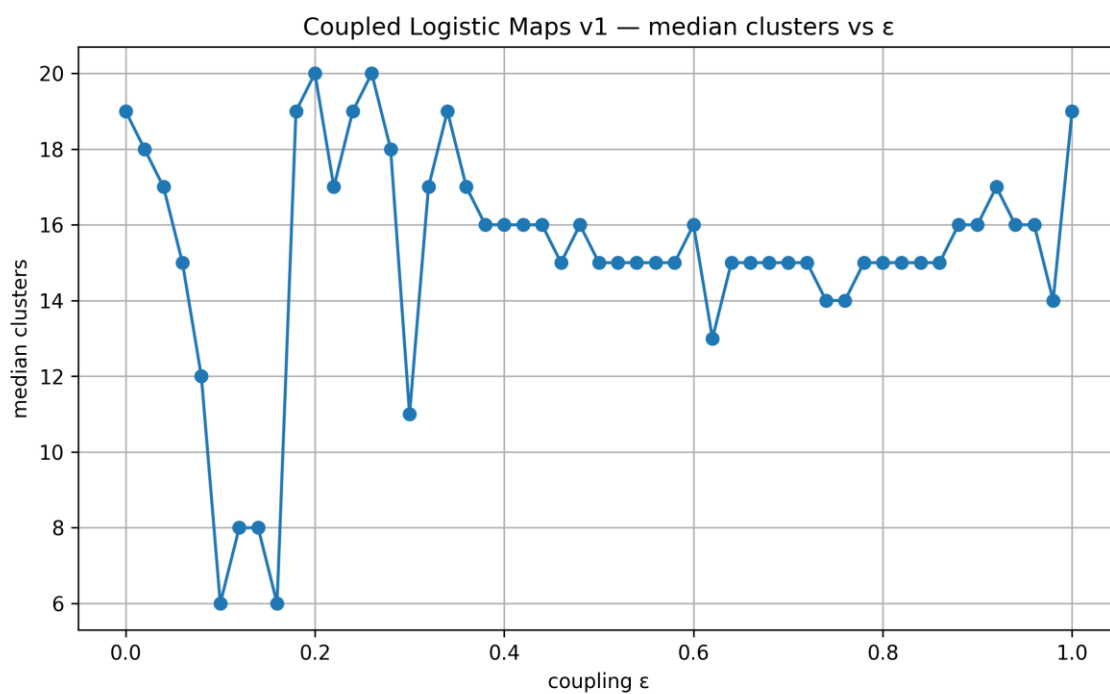


Figure D6. Robustness behavior under normalization and endpoint exclusion scans.

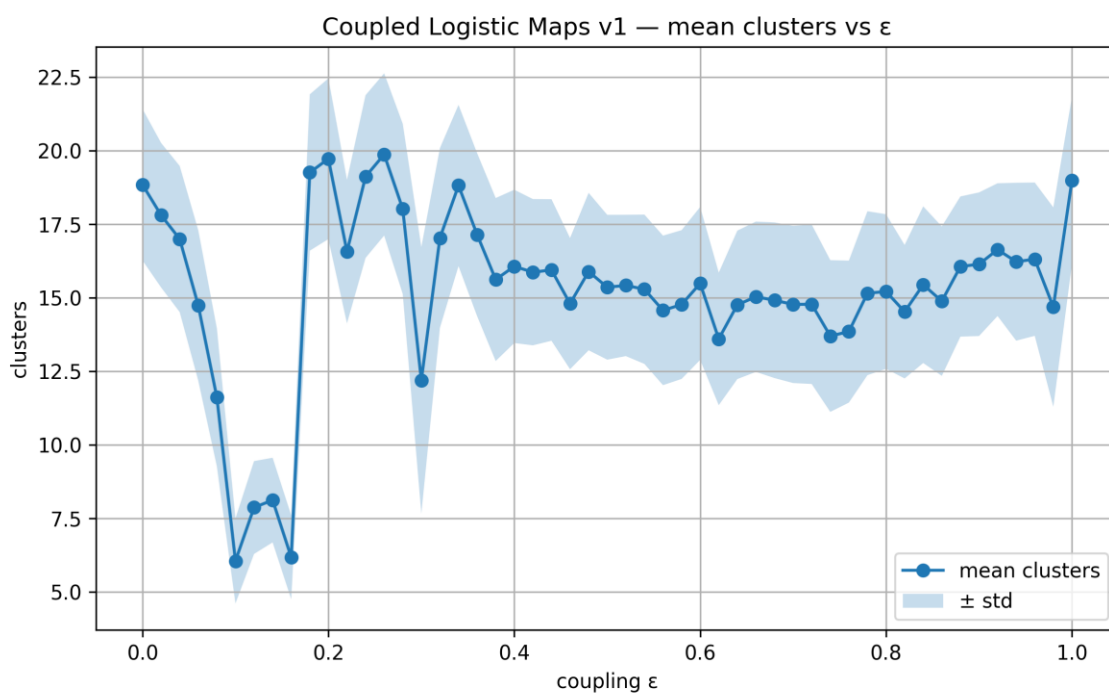


Figure D7. Aggregate distributed organization landscape across coupled map lattice regimes.

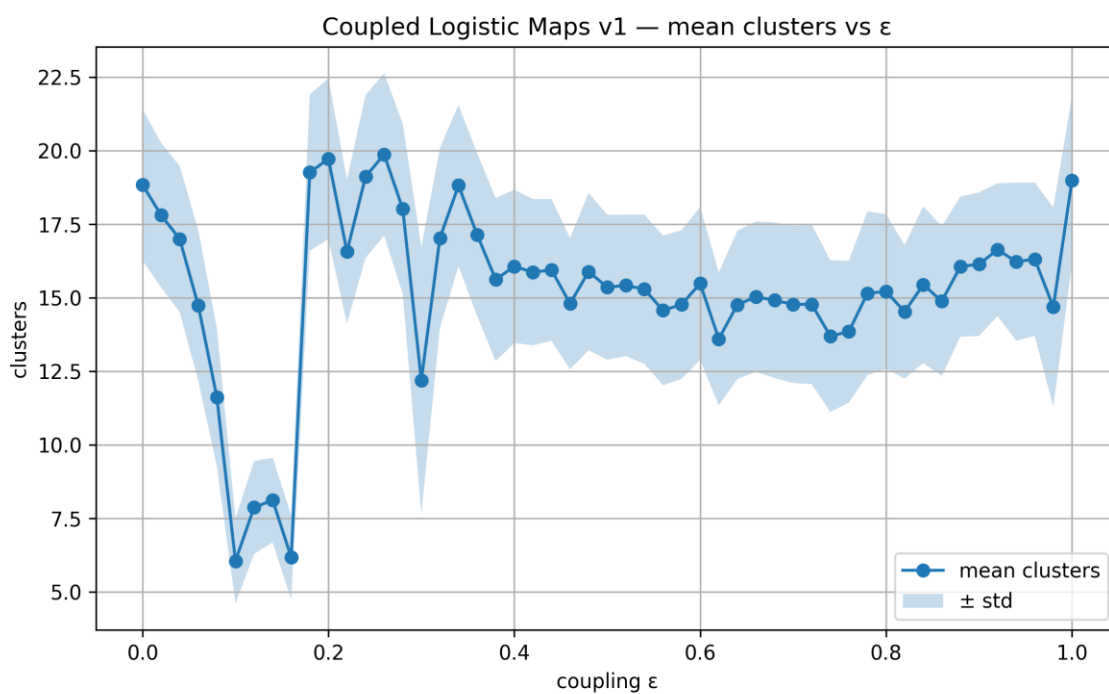


Figure D8. Distributed organization persistence landscape across coupling regimes.

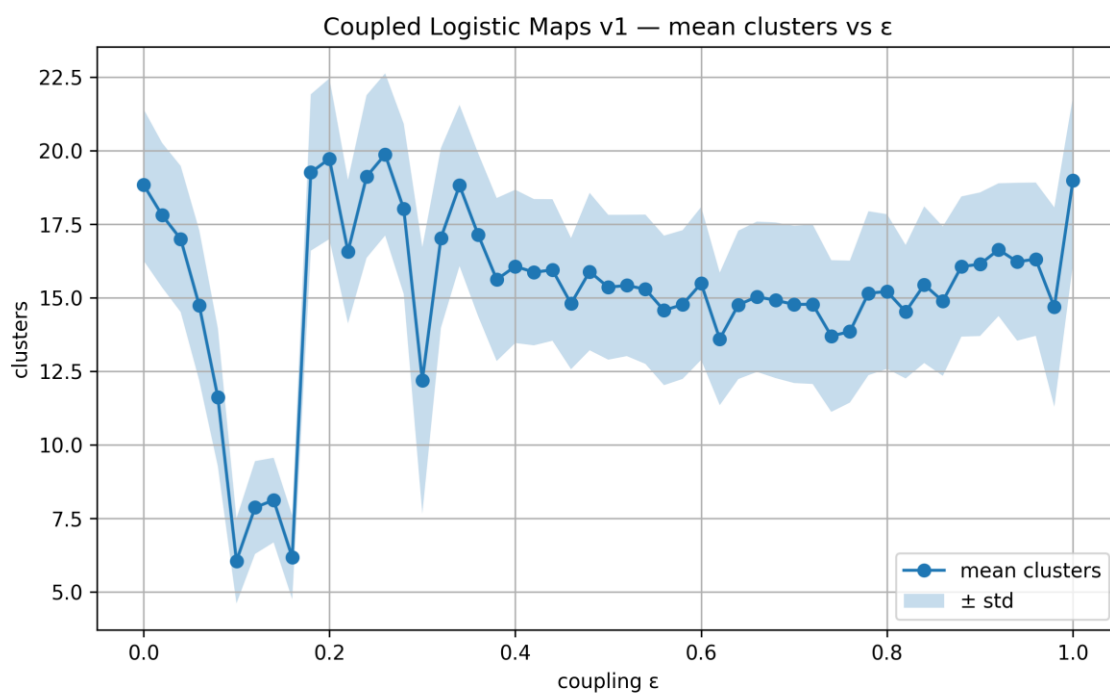


Figure D9. Comparative persistence response under original and shuffled conditions.

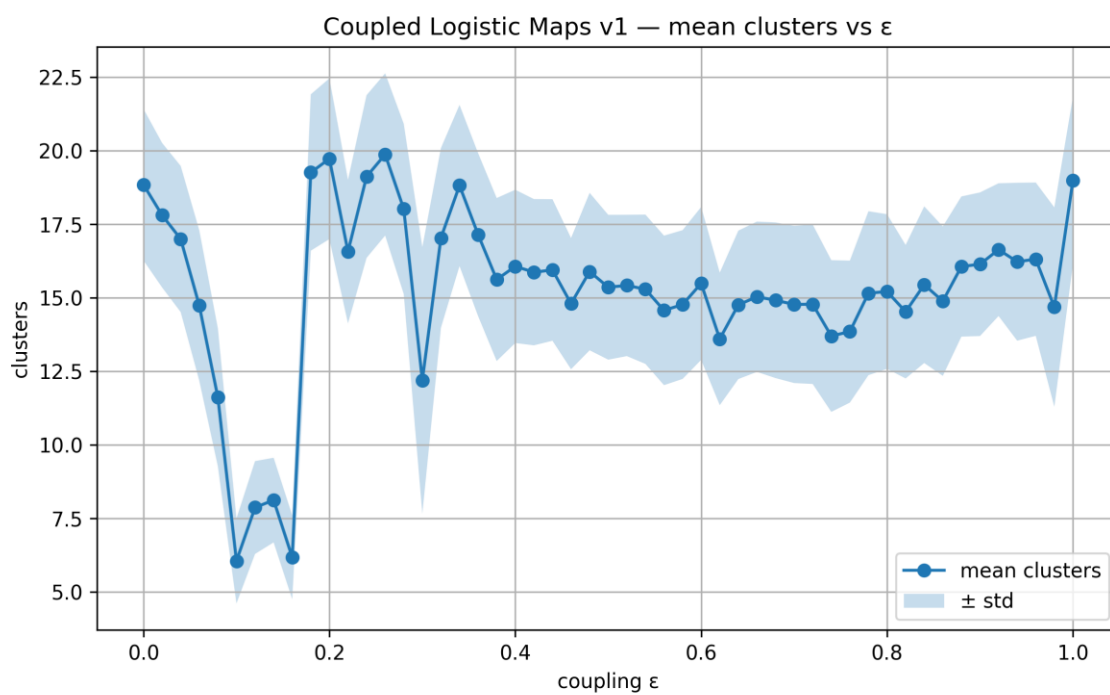


Figure D10. Summary visualization of distributed organization persistence across coupling regimes.

D10. Supplementary Closing Statement

The present supplement extends the robustness and distributed organization layers of the main manuscript. The assembled evidence supports the emergence of compatibility-dependent organization signatures in distributed nonlinear lattice dynamics under intermediate coupling regimes.